

Notes: Grossman & Stiglitz (AER, 1980)

On the Impossibility of Informationally Efficient Markets

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1 Big picture

- **Question.** Can a competitive market be perfectly informationally efficient when information is costly to acquire?
- **Main answer.** No. If prices fully reveal all costly information, informed traders cannot earn a return on becoming informed. But if nobody becomes informed, prices cannot reveal the information. This creates the Grossman–Stiglitz paradox.
- **Core idea.** Markets can be in equilibrium only with an **equilibrium degree of disequilibrium**: prices are informative, but not perfectly informative.
- **Key friction.** Information is privately costly. Traders who pay to observe the signal must be compensated through trading profits.
- **How prices reveal information.** Informed traders buy more when their signal says the risky asset payoff is high and sell more when the signal says it is low. Their trading moves the price, so uninformed traders infer part of the signal from the price.
- **Why revelation is incomplete.** Noisy supply prevents uninformed traders from distinguishing price movements caused by information from price movements caused by supply shocks.
- **Endogenous information acquisition.** The fraction of informed traders, λ , is determined by an information-market equilibrium: expected utility from being informed, net of information cost, equals expected utility from remaining uninformed.
- **Main comparative statics.** Prices become more informative when information is more precise, information is cheaper, or traders are less risk averse. More supply noise increases the equilibrium number of informed traders, but in the CARA-normal example it does not change equilibrium price informativeness because λ adjusts.
- **No full efficiency.** If there is no noise, or if informed traders receive perfect information, prices fully reveal the signal whenever any traders are informed. Then informed traders do not want to pay for information, while if no one is informed some trader wants to become informed. Equilibrium fails.
- **Thin markets.** When almost nobody is informed or almost everybody is informed, beliefs become too similar and speculative trade becomes thin. Markets that reveal too much information can lose the very disagreement that sustains them.
- **Economic takeaway.** The efficient markets hypothesis cannot literally mean that prices fully reflect all costly information. A viable market must leave enough informational rents to reward information production.

2 Model primitives and measurement of information

2.1 Two assets and the risky payoff

There are two securities:

- a risk-free asset with gross return R ;
- a risky asset with random payoff u .

The risky payoff is decomposed as

$$u = \theta + \varepsilon, \tag{1}$$

where θ is observable at cost c and ε is not observable. In the paper's interpretation, θ is the costly signal and ε is residual payoff risk.

Interpretation. If θ were freely observable, prices could aggregate and transmit it without compensating anyone. The paradox appears because observing θ costs real resources.

2.2 Informed and uninformed traders

- **Informed traders** pay cost c and observe θ .
- **Uninformed traders** do not observe θ directly. They observe only the price P .
- Both types have rational expectations: uninformed traders understand the equilibrium relation between prices and returns.
- The fraction of informed traders is $\lambda \in [0, 1]$.

Interpretation. The model is not about irrational uninformed traders. They use prices rationally. The key is that prices are noisy signals, not complete revelations.

2.3 Noisy supply

Let x denote the per capita supply of the risky asset. Uninformed traders do not observe x .

- A high price may mean that informed traders observed a high θ .
- Or it may mean that aggregate supply x is low.
- Because uninformed traders cannot separately observe x , price contains noise.

Interpretation. The noisy supply shock is what prevents the price from perfectly revealing the informed signal. Without this noise, the price would fully reveal θ whenever $\lambda > 0$.

2.4 CARA-normal assumptions

The tractable example assumes:

- exponential utility,

$$V(W) = -e^{-aW}, \quad a > 0, \tag{2}$$

where a is absolute risk aversion;

- θ , ε , and x are jointly normal with appropriate independence assumptions;
- $E[\varepsilon] = 0$, $E[\theta\varepsilon] = 0$, and

$$\text{Var}(u \mid \theta) = \text{Var}(\varepsilon) = \sigma_\varepsilon^2. \tag{3}$$

Interpretation. CARA utility and normal shocks make individual demands linear and allow the price equilibrium to be solved explicitly.

2.5 Meaning of informativeness

The equilibrium price turns out to be informationally equivalent to a noisy signal

$$w_\lambda(\theta, x) = \theta - \frac{a\sigma_\varepsilon^2}{\lambda}(x - \mathbb{E}[x]), \quad \lambda > 0. \quad (4)$$

The conditional variance is

$$\text{Var}(w_\lambda \mid \theta) = \left(\frac{a\sigma_\varepsilon^2}{\lambda} \right)^2 \text{Var}(x). \quad (5)$$

Interpretation. A price is more informative when $\text{Var}(w_\lambda \mid \theta)$ is smaller. Prices become more informative when there are more informed traders, when traders are less risk averse, when the residual payoff risk is smaller, or when the noisy supply shock is smaller.

2.6 Why the model has no empirical data section

This paper is theoretical. There are no empirical data, no regression sample, and no tables. The analogue of data measurement is the paper's formal measurement of information: how much θ can be inferred from P , or equivalently from w_λ .

Interpretation. The key measured object is not a dataset variable but the informativeness of prices, summarized by conditional variance and squared correlation.

3 Models and theoretical specifications

3.1 Individual wealth and budget constraint

Let M_i be the risk-free asset held by trader i , and let X_i be the risky asset position. The budget constraint is

$$PX_i + M_i = W_{0i}. \quad (6)$$

End-of-period wealth is

$$W_{1i} = RM_i + uX_i. \quad (7)$$

Interpretation. Since u is random, portfolio choice depends on the conditional distribution of u given the trader's information.

3.2 Demand of informed traders

An informed trader observes θ . With CARA-normal preferences, maximizing expected utility yields

$$X_I(P, \theta) = \frac{\theta - RP}{a\sigma_\varepsilon^2}. \quad (8)$$

Interpretation. Informed demand increases when the signal θ is high relative to the price. Demand is more aggressive when risk aversion a is low or residual risk σ_ε^2 is low.

3.3 Demand of uninformed traders

Uninformed traders observe the price and infer from it. Their demand is

$$X_U(P; P^*) = \frac{E[u \mid P^* = P] - RP}{a \text{Var}(u \mid P^* = P)}. \quad (9)$$

Interpretation. The uninformed are rational Bayesian learners. Their problem is that the price is a noisy signal of θ .

3.4 Equilibrium price for a given fraction informed

For a fixed fraction λ of informed traders, an equilibrium price function $P_\lambda(\theta, x)$ clears the market:

$$\lambda X_I(P_\lambda(\theta, x), \theta) + (1 - \lambda)X_U(P_\lambda(\theta, x); P_\lambda) = x. \quad (10)$$

The paper proves that, for $\lambda > 0$, the equilibrium price is a linear function of w_λ :

$$P_\lambda(\theta, x) = a_1 + a_2 w_\lambda(\theta, x), \quad a_2 > 0. \quad (11)$$

Interpretation. The price itself is not arbitrary. It is a sufficient statistic for the noisy signal w_λ , which combines true information θ and supply noise x .

3.5 Information-market equilibrium

Let W_I^λ be final wealth if the trader pays to become informed, and W_U^λ final wealth if the trader remains uninformed:

$$W_I^\lambda = R(W_{0i} - c) + [u - RP_\lambda(\theta, x)]X_I(P_\lambda(\theta, x), \theta), \quad (12)$$

$$W_U^\lambda = RW_{0i} + [u - RP_\lambda(\theta, x)]X_U(P_\lambda(\theta, x); P_\lambda). \quad (13)$$

The key expected-utility ratio is

$$\frac{E V(W_I^\lambda)}{E V(W_U^\lambda)} = e^{ac} \sqrt{\frac{\text{Var}(u | \theta)}{\text{Var}(u | w_\lambda)}} \equiv \gamma(\lambda). \quad (14)$$

For an interior equilibrium,

$$\gamma(\lambda) = 1. \quad (15)$$

Interpretation. The informed must be just indifferent to paying for information. If information rents are too high, more traders become informed. If information rents are too low, fewer traders become informed.

3.6 Why the sign of the utility ratio can look paradoxical

Because exponential utility is written as a negative function, a larger numerical value of the ratio does not mean informed traders are better off. The paper notes that although $\gamma(\lambda)$ is increasing in λ , expected utility of the informed declines relative to that of the uninformed as λ rises.

Interpretation. More informed traders make prices more informative and reduce each informed trader's informational advantage.

3.7 Price informativeness in closed form

Define

$$m = \left(\frac{a\sigma_\varepsilon^2}{\lambda} \right)^2 \frac{\sigma_x^2}{\sigma_\theta^2}, \quad n = \frac{\sigma_\theta^2}{\sigma_\varepsilon^2}. \quad (16)$$

The squared correlation between price and the signal is

$$\rho_{\theta p}^2 = \frac{1}{1+m}. \quad (17)$$

The ratio of information quality between informed and uninformed traders is

$$\frac{\text{Var}(u | \theta)}{\text{Var}(u | w_\lambda)} = \frac{1+m}{1+m+nm} = \left(1 + \frac{nm}{1+m} \right)^{-1}. \quad (18)$$

At an interior equilibrium,

$$m = \frac{e^{2ac} - 1}{1 + n - e^{2ac}}, \quad \text{and} \quad 1 - \rho_{\theta p}^2 = \frac{e^{2ac} - 1}{n}. \quad (19)$$

Interpretation. Price informativeness is pinned down by information cost c , information quality n , and risk aversion a . The endogenous fraction informed adjusts to sustain this informativeness.

3.8 Comparative statics

The paper's Theorem 4 implies the following for an interior equilibrium:

- Price informativeness rises if information quality n rises.
- Price informativeness rises if information cost c falls.
- Price informativeness rises if risk aversion a falls.
- Price informativeness is unchanged by a change in supply noise σ_x^2 because the equilibrium fraction informed adjusts.
- The fraction informed rises if supply noise rises.
- The fraction informed rises if information cost falls.
- The effect of information quality on the fraction informed is ambiguous: better information raises the value of becoming informed, but it also makes prices more informative for the uninformed.

Interpretation. The market endogenously balances information production against information leakage through prices.

3.9 No equilibrium with fully revealing prices

If there is no supply noise, or if informed traders receive perfect information, then prices fully reveal the signal whenever $\lambda > 0$. The paper's Theorem 5 shows that in this case an overall equilibrium need not exist.

- If $\lambda > 0$, informed traders pay a cost but price already reveals their information, so each wants to stop paying for information.
- If $\lambda = 0$, price reveals nothing, so someone wants to become informed if the information cost is low enough.

Interpretation. Exact informational efficiency destroys the private incentive to acquire information. This is the paradox.

3.10 Thin speculative markets

Trade exists because agents differ in endowments, preferences, or beliefs. This paper emphasizes differences in beliefs. As $\lambda \rightarrow 0$ or $\lambda \rightarrow 1$, beliefs become more homogeneous, and the mean and variance of net informed trade go to zero in the limiting cases studied by the paper.

Interpretation. A market that reveals too much information can become too thin to operate. The information-revelation role of markets is self-limiting.

4 Theoretical design and credibility of the argument

4.1 What makes the argument work

- The model makes information acquisition endogenous.
- Prices aggregate information through informed traders' demands.
- Supply noise prevents prices from being perfectly revealing.
- Rational uninformed traders learn from the price but cannot fully infer the signal.
- Information-market equilibrium requires informed and uninformed traders to earn equal expected utility net of information costs.

Interpretation. The paper's credibility comes from closing both markets: the asset market and the information market.

4.2 Why noisy supply is essential

Without noisy supply, prices would fully reveal the informed signal. The model therefore uses supply noise as a formal way to ensure that price movements are not perfectly interpretable.

Interpretation. Noise is not a nuisance in the model. Noise is what makes costly information acquisition compatible with equilibrium.

4.3 Why CARA-normal assumptions matter

- CARA utility eliminates wealth effects in demand.
- Normality makes conditional expectations linear.
- These assumptions deliver a price that is linear in w_λ .
- They allow explicit expressions for information value and price informativeness.

Interpretation. The paper proves the conjectures in a tractable benchmark. The logic is broader than the exact distributional assumptions, but the formal proof is tied to them.

4.4 What the paper says about the efficient markets hypothesis

The paper does not reject the idea that prices convey information. It rejects the literal version that prices fully reflect all available costly information.

- If prices reveal no information, uninformed traders learn nothing and information acquisition is valuable.
- If prices reveal all information, informed traders earn no compensation for acquiring it.
- Equilibrium lies between these extremes.

Interpretation. The efficient market is not perfectly efficient. It is efficient enough to aggregate some information, but inefficient enough to reward information production.

4.5 Why market completeness is problematic

The paper argues that if there were enough securities to span all states, additional prices could reveal more information. But if all information is revealed, the same no-equilibrium logic applies.

Interpretation. Market incompleteness and transaction costs can help sustain equilibrium by preventing prices from revealing everything.

4.6 What the paper cannot fully prove

- The paper does not provide a fully general existence proof for all preferences and distributions.
- The main formal results are proven in the CARA-normal example.
- The welfare properties of equilibrium are not fully analyzed in the paper.
- The model abstracts from many real-world features: strategic information acquisition, multiple signals, institutions, delegated asset management, and dynamic trading.
- It does not empirically test whether real markets are at the equilibrium degree of disequilibrium.

Interpretation. The contribution is conceptual and theoretical: it shows that perfect informational efficiency is internally inconsistent when information is costly.

5 Tables and figures

The paper has no empirical tables and one main figure. The figure below is directly cropped from the original paper and placed beside the reading notes to keep the layout compact.

5.1 Figure 1: Determination of the equilibrium fraction informed

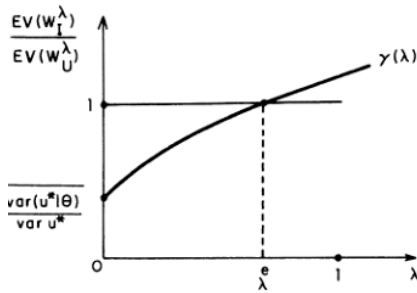


FIGURE 1

Original paper Figure 1: equilibrium λ .

Read:

- Horizontal axis: the fraction of informed traders, λ .
- Vertical axis: the expected-utility ratio $\gamma(\lambda)$.
- The curve crosses one at the interior equilibrium λ^e .
- As λ rises, price becomes more informative and the informed advantage falls.
- Equilibrium requires informed and uninformed traders to be indifferent after information cost.

Economic message. The figure is the core of the paper. Information production survives only when prices are informative enough for markets to function, but not so informative that informed traders lose their private reward.

5.2 Equation block: price as a noisy signal

Read:

- The equilibrium price is informationally equivalent to

$$w_\lambda = \theta - \frac{a\sigma_\varepsilon^2}{\lambda}(x - \mathbb{E}x).$$

- The noisy component is increasing in risk aversion a , residual payoff variance σ_ε^2 , and supply noise $x - E x$.
- The noisy component is decreasing in the fraction informed λ .

Economic message. The price system transmits a distorted version of the informed signal. This distortion is exactly what creates compensation for informed traders.

5.3 Equation block: equilibrium informativeness

Read:

- Define $n = \sigma_\theta^2 / \sigma_\varepsilon^2$ as information quality.
- The interior equilibrium implies

$$1 - \rho_{\theta p}^2 = \frac{e^{2ac} - 1}{n}.$$

- Lower information cost c means a more informative price.
- Higher information quality n means a more informative price.
- Lower risk aversion a means informed traders trade more aggressively, so price becomes more informative.

Economic message. The exact degree of price informativeness is endogenous. It is determined by the cost and quality of information and by the aggressiveness of informed traders.

5.4 No-table note

There are no empirical tables in this article. The central quantitative objects are theoretical: λ , w_λ , $\gamma(\lambda)$, m , n , and $\rho_{\theta p}^2$.

Economic message. For this paper, the correct substitute for table-by-table reading is theorem-by-theorem reading.

6 Short summary

- The paper asks whether perfectly informationally efficient markets are possible when information is costly.
- It argues that they are not: if prices fully reveal information, no one will pay to become informed.
- The model has a safe asset, a risky asset, informed traders who observe a costly signal, and uninformed traders who infer from price.
- The risky payoff is $u = \theta + \varepsilon$.
- The fraction of informed traders, λ , is endogenous.
- Supply noise prevents price from perfectly revealing the informed signal.
- For a given λ , the equilibrium price is a linear function of a noisy signal w_λ .
- As more traders become informed, the price becomes more informative.
- As the price becomes more informative, the value of becoming informed falls.
- The information-market equilibrium equates expected utility from being informed and uninformed.
- Interior equilibrium requires $\gamma(\lambda) = 1$.
- Price informativeness rises when information quality rises, information cost falls, or risk aversion falls.
- More supply noise induces more traders to become informed, offsetting the reduction in price informativeness in the CARA-normal case.
- If there is no supply noise, prices fully reveal information whenever any traders are informed, so no positive-information equilibrium exists.
- If nobody is informed, prices reveal nothing, so some trader wants to acquire information if the cost is low enough.
- Therefore perfect informational efficiency is inconsistent with costly information acquisition.
- Speculative markets become thin when beliefs are too similar, especially when almost everyone or almost no one is informed.
- Market incompleteness and transaction costs can help sustain equilibrium because they prevent all information from being revealed.
- The paper redefines efficient markets: prices can be informative, but they cannot fully reflect all costly information.

7 Conclusion

7.1 Core conclusion

Grossman and Stiglitz show that perfectly informationally efficient markets are impossible when information is costly. Prices must reveal some information, otherwise uninformed traders cannot learn from markets. But prices cannot reveal all information, otherwise informed traders cannot recover the cost of becoming informed.

7.2 Economic intuition

Information production requires rents. Informed traders collect these rents by trading against less-informed traders. But their trading makes the price more informative, which transfers part of their information to the uninformed. Equilibrium occurs when just enough information is revealed through prices that the informed are exactly compensated for the cost of acquiring information.

7.3 Takeaway

The efficient markets hypothesis should not be interpreted as perfect price revelation. A market with costly information must have some noise, some residual disagreement, and some trading profits for informed investors. The central lesson is that informational efficiency and incentives to produce information are in tension.